

COMPUTER NETWORKS

AKTU Exam — Complete Revision Notes

All 10M / 5M Topics Covered

1. OSI REFERENCE MODEL (7 Layers)

The **Open Systems Interconnection (OSI)** model is a conceptual framework that standardises network communication into **7 layers**. Each layer provides services to the layer above and uses services of the layer below.

Layer #	Layer Name	Function	Protocols / Examples
7	Application	Network services to end-user apps; UI	HTTP, FTP, SMTP, DNS, Telnet
6	Presentation	Data translation, encryption, compression	SSL/TLS, JPEG, MPEG, ASCII
5	Session	Session management; dialog control	NetBIOS, RPC, PPTP
4	Transport	End-to-end delivery; segmentation; flow/error control	TCP, UDP
3	Network	Logical addressing; routing; packet forwarding	IP, ICMP, OSPF, RIP, ARP
2	Data Link	Frame framing; MAC addressing; error detection	Ethernet, PPP, HDLC, Wi-Fi
1	Physical	Bit transmission over physical medium	RS-232, USB, Hubs, DSL

OSI Model Diagram

SENDER RECEIVER

[illegible]

Layer-wise PDU Names

- Layer 7/6/5 → Data
- Layer 4 (Transport) → Segment (TCP) / Datagram (UDP)
- Layer 3 (Network) → Packet
- Layer 2 (Data Link) → Frame
- Layer 1 (Physical) → Bits

OSI vs TCP/IP Model Comparison

Feature	OSI Model	TCP/IP Model
Layers	7	4

Layer Names	Physical, Data Link, Network, Transport, Session, Presentation, Application	Network Access, Internet, Transport, Application
Development	Developed before protocols	Protocols defined first
Usage	Conceptual/Reference only	Practical / Widely used
Reliability	Defined in Transport layer	End-to-end responsibility
Protocols	Generic	TCP, IP, UDP, HTTP, FTP...
Session & Pres. Layers	Separate layers	Merged into Application
Physical & Data Link	Separate layers	Merged into Network Access

2. TCP vs UDP — Detailed Comparison

TCP Header Format (20 bytes minimum)

Field	Size (bits)	Purpose
Source Port	16	Sender port number
Destination Port	16	Receiver port number
Sequence Number	32	Order of data bytes
Acknowledgement No.	32	Next expected byte
Header Length (HLEN)	4	Size of TCP header
Reserved	6	Future use (set to 0)
Control Flags (URG/ACK/PSH/RST/SYN/FIN)	6	Connection control
Window Size	16	Flow control buffer size
Checksum	16	Error detection
Urgent Pointer	16	Points to urgent data
Options + Padding	Variable	Optional extensions

UDP Header Format (8 bytes fixed)

Field	Size (bits)	Purpose
Source Port	16	Sender port
Destination Port	16	Receiver port
Length	16	Header + Data length
Checksum	16	Error detection (optional in IPv4)

TCP vs UDP Comparison Table

Parameter	TCP	UDP
Full Form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable — guarantees delivery	Unreliable — no guarantee
Ordering	Maintains order (sequence numbers)	No ordering
Error Control	Yes (checksum + retransmission)	Only checksum
Flow Control	Yes (sliding window)	No
Congestion Control	Yes	No
Header Size	20–60 bytes	8 bytes (fixed)
Speed	Slower (overhead)	Faster (low overhead)
Broadcasting	Not supported	Supported
Use Cases	HTTP, FTP, SMTP, SSH, Telnet DNS, DHCP, VoIP, Video Streaming, Gaming	

TCP 3-Way Handshake

- **Step 1 — SYN:** Client sends SYN (Synchronize) segment to server. $SEQ = x$
- **Step 2 — SYN-ACK:** Server replies with SYN+ACK. $SEQ = y$, $ACK = x+1$
- **Step 3 — ACK:** Client sends ACK. $ACK = y+1$. Connection established.

Connection Termination: 4-way FIN handshake (FIN → ACK → FIN → ACK)

3. IP ADDRESSING — Classes + Subnetting

IP Address Classes

Class	Range (1st Octet)	Default Mask	Networks	Hosts/Network	Use
A	1 – 126	255.0.0.0 (/8)	126	16,777,214	Large organisations
B	128 – 191	255.255.0.0 (/16)	16,384	65,534	Medium networks
C	192 – 223	255.255.255.0 (/24)	2,097,152	254	Small networks
D	224 – 239	N/A	Multicast	—	Multicast groups
E	240 – 255	N/A	Experimental	—	Research only

• Loopback: **127.0.0.1** • Private A: 10.x.x.x • Private B: 172.16–31.x.x • Private C: 192.168.x.x

Subnetting Numerical: 200.1.2.0 → 5 Subnets

Given: Network **200.1.2.0** (Class C, default mask 255.255.255.0)

Step 1: Find minimum subnet bits $\geq \log_2(5) \rightarrow$ need **3 bits** ($2^3 = 8$ subnets)

Step 2: New Subnet Mask = 255.255.255.224 (borrowing 3 bits from host $\rightarrow$ /27)

Step 3: Block size = $256 - 224 = 32$

Each subnet has: **32 – 2 = 30 usable hosts**

Subnet #	Network Address	Subnet Mask	Broadcast Address	Host Range
1	200.1.2.0	255.255.255.224	200.1.2.31	200.1.2.1 – 200.1.2.30
2	200.1.2.32	255.255.255.224	200.1.2.63	200.1.2.33 – 200.1.2.62
3	200.1.2.64	255.255.255.224	200.1.2.95	200.1.2.65 – 200.1.2.94
4	200.1.2.96	255.255.255.224	200.1.2.127	200.1.2.97 – 200.1.2.126
5	200.1.2.128	255.255.255.224	200.1.2.159	200.1.2.129 – 200.1.2.158

Subnet Address: 180.25.21.x with mask 255.255.192.0

Mask: 255.255.192.0 $\rightarrow$ /18 | Block size in 3rd octet = $256 - 192 = 64$

3rd octet of IP = 21 $\rightarrow$ Floor($21/64$) $\times 64 = 0 \rightarrow$ **Network Address = 180.25.0.0**

Broadcast = 180.25.63.255 | Host Range: 180.25.0.1 – 180.25.63.254

4. CRC NUMERICAL — 1101011011 ÷ 10011

Message (M): 1101011011 | Generator (G): 10011 (degree = 4)

Step 1: Append 4 zeros → 11010110110000

Step 2: Modulo-2 Division (XOR)

11010110110000 ÷ 10011

10011

■■■■■■■■■■■■■■■■■■■■

10011

10011

■■■■■■■■

00001011

10011

■■■■■■■■

10001

10011

■■■■■■■■

00100 ← Remainder (CRC)

CRC / Remainder = 0100

Transmitted Frame = 1101011011 + 0100 = 11010110110100

At receiver: divide received frame by same generator — if remainder = 0000, NO error.

5. SLIDING WINDOW — Go-Back-N (GBN)

Key Formulae

- **Efficiency (Stop-and-Wait):** $\eta = 1 / (1 + 2a)$ where $a = T_p/T_t$
- **Efficiency (GBN, $W \leq 2a+1$):** $\eta = W / (1 + 2a)$
- **Efficiency (GBN, $W \geq 2a+1$):** $\eta = 1$
- **Window Size (GBN):** $W \leq 2^n - 1$ (n = sequence number bits)
- **Window Size (SR):** $W \leq 2^{n-1}$
- **Throughput:** $= W \times \text{Frame_size} / (1 + 2a) \times T_t$

GBN vs Selective Repeat

Feature	Go-Back-N (GBN)	Selective Repeat (SR)
On Error	Re-sends errored + all subsequent	Re-sends only errored frame
Sender Window	$2^n - 1$	2^{n-1}
Receiver Window	1	2^{n-1}
Buffer (Receiver)	Not required	Required
Complexity	Lower	Higher
Efficiency	Lower (wasted retransmits)	Higher

6. ROUTING — DVR & Dijkstra

Distance Vector Routing (DVR / Bellman-Ford)

- Each router maintains a table of (destination, distance, next-hop).
- Routers share their table with **direct neighbours** periodically.
- Update rule: if $d(X,Y) + d(Y,Z) < d(X,Z) \rightarrow$ update entry.
- Converges slowly; suffers from **Count-to-Infinity problem**.
- Fix: Split horizon, Poison reverse, Hold-down timers.
- Protocol: **RIP** (max hop = 15)

Count-to-Infinity Problem

When a link fails, routers keep incrementing distance believing a path exists through each other. Distance grows until it hits infinity (16 in RIP). Cause: slow convergence of DVR.

Dijkstra's Algorithm (Link State)

- Each router floods LSA (Link State Advertisement) to all routers.
- Every router builds complete topology map.
- Dijkstra's SPF algorithm computes shortest path tree.
- Faster convergence; no count-to-infinity.
- Protocol: **OSPF**

Dijkstra Example (Typical AKTU format)

Given graph: A–B=2, A–C=4, B–C=1, B–D=7, C–D=3. Find shortest path from A.

Step	Visited	Dist(A)	Dist(B)	Dist(C)	Dist(D)
0	{A}	0	∞	∞	∞
1	{A,B}	0	2	4	∞
2	{A,B,C}	0	2	3	6
3	{A,B,C,D}	0	2	3	6

Shortest paths: A→B=2, A→C=3 (via B), A→D=6 (via B→C)

7. RSA NUMERICAL (p=7, q=11)

RSA Steps:

- Given: $p = 7$, $q = 11$
- $n = p \times q = 7 \times 11 = 77$
- $\phi(n) = (p-1)(q-1) = 6 \times 10 = 60$
- Choose e : $1 < e < 60$, $\gcd(e, 60) = 1 \rightarrow e = 7$ (since $\gcd(7, 60) = 1$ ✓)
- Find d : $d \times e \equiv 1 \pmod{60} \rightarrow 7d \equiv 1 \pmod{60} \rightarrow d = 43$ ($7 \times 43 = 301 = 5 \times 60 + 1$)
- Public Key = $(e, n) = (7, 77)$
- Private Key = $(d, n) = (43, 77)$

Encryption & Decryption

Let $M = 2$ (plaintext message)

- Encrypt: $C = M^e \bmod n = 2^7 \bmod 77 = 128 \bmod 77 = 51$
- Decrypt: $M = C^d \bmod n = 51^{43} \bmod 77 = 2$ ✓

Security relies on difficulty of factoring large n into p and q .

8. CONGESTION CONTROL — Leaky Bucket & Token Bucket

Leaky Bucket Algorithm

- Analogy: water poured at any rate leaks out at a **constant rate**.
- Incoming packets stored in bucket (queue); output at fixed rate r .
- If bucket full $\rightarrow$ packets **dropped**.
- Smooths bursty traffic into steady stream.
- **Disadvantage**: Cannot handle sudden legitimate bursts.

Token Bucket Algorithm

- Tokens generated at rate r tokens/sec; bucket holds max B tokens.
- To send a packet $\rightarrow$ must consume a token.
- Allows **bursts** up to bucket size B .
- More flexible than Leaky Bucket.
- **Advantage**: Burst traffic allowed (up to B packets at once).

Feature	Leaky Bucket	Token Bucket
Output Rate	Constant	Variable (up to burst limit)
Burst Handling	Does not allow	Allows up to B tokens
Packet Drop	When bucket full	When no tokens available
Usage	Traffic shaping	Traffic policing & shaping

9. FTP + SMTP + DNS

FTP (File Transfer Protocol) — Port 20/21

- Port 21 = Control connection (commands); Port 20 = Data connection.
- Works in **Active Mode** (server initiates data) and **Passive Mode** (client initiates both).
- Uses two separate TCP connections.
- Commands: USER, PASS, LIST, GET, PUT, QUIT.

SMTP (Simple Mail Transfer Protocol) — Port 25

- Used to **send** email from client to server and between mail servers.
- Works over TCP, port **25** (or 587 for submission).
- Commands: HELO, MAIL FROM, RCPT TO, DATA, QUIT.
- To **retrieve** email: POP3 (port 110) or IMAP (port 143).

DNS (Domain Name System) — Port 53

- Translates domain names (www.google.com) to IP addresses.
- Uses UDP port 53 (TCP for zone transfers).
- Hierarchical: Root → TLD (.com) → Authoritative → Local DNS.
- Record types: A (IPv4), AAAA (IPv6), MX (mail), CNAME (alias), NS, PTR.
- Iterative vs Recursive query resolution.

10. SHORT ANSWER TOPICS (2M / 5M)

Bit Rate vs Baud Rate

Bit Rate: Number of bits transmitted per second (bps). Measures data throughput.

Baud Rate: Number of signal changes (symbols) per second. 1 baud can carry multiple bits.

Relation: Bit Rate = Baud Rate $\times \log_2(M)$, where M = number of signal levels.

Piggybacking

Technique where **acknowledgement (ACK)** is attached/carried along with an outgoing data frame in the reverse direction, instead of sending a separate ACK frame. Improves efficiency by reducing overhead.

Point-to-Point vs Point-to-Multipoint

Feature	Point-to-Point	Point-to-Multipoint
Connection	One sender ↔ One receiver	One sender → Multiple receivers
Example	Leased lines, PPP links	Wi-Fi, Cable TV, Broadcast
Bandwidth	Dedicated	Shared

ICMP (Internet Control Message Protocol)

- Works at **Network Layer** (Layer 3); part of IP suite.
- Used for **error reporting and diagnostics**, NOT data transfer.
- Messages: Echo Request/Reply (ping), Destination Unreachable, Time Exceeded, Redirect.
- Used by tools: **ping** and **traceroute**.

Port Addressing — Well-Known Ports

Port	Protocol	Service
20/21	FTP	File Transfer
22	SSH	Secure Shell
23	Telnet	Remote Login
25	SMTP	Email Send
53	DNS	Domain Name
67/68	DHCP	IP Assignment
80	HTTP	Web (unsecure)
110	POP3	Email Receive
143	IMAP	Email (advanced)
443	HTTPS	Secure Web

ARP and RARP

- **ARP (Address Resolution Protocol):** Maps IP address → MAC address. Sends broadcast to find MAC for known IP.
- **RARP (Reverse ARP):** Maps MAC address → IP address. Used by diskless machines to obtain IP at boot.

Symmetric vs Asymmetric Encryption (One-liner each)

Symmetric: Same key used for both encryption and decryption. (e.g., AES, DES) — Fast but key-sharing problem.

Asymmetric: Different keys — Public key encrypts, Private key decrypts. (e.g., RSA) — Slow but secure.

QoS (Quality of Service) Parameters

- **Bandwidth:** Maximum data rate available.
- **Delay (Latency):** Time to deliver a packet from source to destination.
- **Jitter:** Variation in packet delay. Critical for real-time applications.
- **Packet Loss:** Percentage of packets dropped.
- **Reliability:** Probability of successful delivery.
- **Throughput:** Actual data successfully transferred per second.
- **Priority:** Preference given to certain traffic types.

QUICK REVISION — Exam Cheatsheet

Topic	Key Point / Formula
OSI Layers	All People Seem To Need Data Processing (7→1)
TCP 3-Way	SYN → SYN-ACK → ACK
Subnetting	Block = 256 – last non-255 octet of mask
CRC	Append (n-1) zeros → XOR divide → remainder = FCS
GBN Window	$W \leq 2^k - 1$
SR Window	$W \leq 2^{k-1}$
RSA n	$n = p \times q$
RSA $\phi(n)$	$(p-1)(q-1)$
RSA public	e: $\gcd(e, \phi(n)) = 1$
RSA private	$d \times e \equiv 1 \pmod{\phi(n)}$
Leaky Bucket	Constant output rate; drops excess
Token Bucket	Variable output; allows bursts up to B
ARP	IP → MAC (broadcast query)
ICMP	Error reporting (ping, traceroute)
DNS port	53 (UDP); 53 (TCP for zone transfer)
SMTP port	25 (server-server), 587 (client-server)
HTTP/HTTPS	80 / 443
Bit Rate formula	Bit Rate = Baud Rate $\times \log_2(M)$
Efficiency (S&W)	$\eta = 1/(1+2a)$
Efficiency (GBN)	$\eta = W/(1+2a)$ if $W < 2a+1$

Best of luck for your AKTU exam! ■